Technology Stack

Date	27 June 2026
Team ID	SWTID-2026-3658
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:
Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.
Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	Streamlit, HTML, CSS	Streamlit was chosen to rapidly develop an interactive and responsive web interface with minimal code. Custom HTML and CSS were used to enhance UI appearance and improve user experience.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, FastAPI	FastAPI was selected because it provides high-performance REST APIs, automatic API documentation through Swagger UI, and easy integration with AI/ML modules.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	JSON Files (history.json, feedback.json)	JSON files were used for lightweight local storage of conversation history and user feedback without requiring a dedicated database server.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Local Deployment using Uvicorn and Streamlit	The application is currently deployed locally for development and demonstration purposes. The architecture can be easily migrated to cloud platforms in future versions.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub	Git and GitHub were used for source code management, version tracking, collaboration, and project backup.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Wikipedia API, Hugging Face Transformers, PyTest, ReportLab	Wikipedia API enables fact verification, Hugging Face Transformers provide NLP capabilities for event analysis, PyTest is used for automated testing, and ReportLab generates downloadable PDF reports.